TradingAI Concepts, Indicators, Calculations and C

Complete current quantitative and decision-intelligence reference.

System reference date: 2026-08-29

Standalone current-state reference for the complete platform through Milestone 78.

1. Price, trend and momentum

- Exponential moving averages and moving-average structure characterize directional alignment and trend persistence across timeframes.
- RSI supplies normalized momentum and overbought/oversold evidence; it is context rather than a standalone trade trigger.
- MACD and histogram behavior contribute crossover, acceleration and momentum-regime evidence.
- ATR normalizes structural distances, breakout buffers, stops, targets, realized excursions and research payoff geometry.
- Multi-horizon returns and trend slopes are used in scanner ranking, trend intelligence, research attribution and contextual conditioning.
- Daily, weekly and monthly trend states are treated as separate but related authorities; transition intelligence tracks changes between states.

2. Structural price intelligence

- Support/resistance levels are consolidated into governed structural levels and zones using strength, confluence, hold behavior and recency.
- Breakout and breakdown logic distinguishes setup, confirmation, retest, continuation and failure rather than treating every move through a level as the same event.
- Failed breakout/breakdown patterns capture trap/reversal mechanisms that are economically distinct from ordinary trend reversal.
- Structural invalidation is used to define trade thesis failure, stop geometry and setup outcome labels.

3. Volume, participation and market structure

- Rolling volume ratios, participation measures and leadership evidence describe whether price movement is broadly supported.
- Dealer-positioning intelligence combines open interest, option Greeks, GEX, Vanna, Charm, volume flow, OI change, wall/flip migration, confidence and conviction.
- Futures/OPEX context contributes pre-open positioning, expiry-related flow and market-structure evidence.
- Relative strength, accumulation and volatility compression remain conditioning evidence where relevant; M77-negative standalone formulations are not revived as automatic alpha.

4. Market breadth, regime and cross-asset context

- Market Overview synthesizes breadth, health, trend/momentum, regime, sentiment, sector rotation, dealer context, volatility/options conditions, liquidity/participation, cross-asset confirmation and risk.
- Pairwise/index/sector correlation and dispersion regimes identify concentration, diversification and systemic-risk conditions.
- Sector membership and constituent breadth provide cross-sectional context for symbol-level decisions.
- An internally derived sentiment ensemble combines multiple market components rather than relying on one external sentiment feed.

5. Option pricing, volatility and relative value

- Option valuation combines market quotes, volatility measures, surface/skew/term structure, event context and relative-value comparisons.
- Implied volatility is compared with realized/forecast volatility and expected movement to identify expensive/cheap volatility conditions.
- Volatility surface analytics evaluate strike- and expiry-dependent pricing rather than assuming one scalar IV.
- Event mispricing, dealer-flow mispricing and relative-value divergence contribute to governed valuation scores.
- Greeks (delta, gamma, theta, vega and higher-order dealer Greeks where relevant) are evaluated both at contract and portfolio level.

6. Probability and distribution analytics

- Probability of profit, target probabilities, expected return and expected value are treated as distributions rather than deterministic forecasts.
- Monte Carlo/scenario analysis evaluates bull, bear, crash, gap, IV expansion/crush and decay conditions.
- Tail-risk analytics include VaR/expected-shortfall style concepts, adverse-tail distributions and stress surfaces.
- Probability calibration compares predicted probabilities with realized outcomes and supports reliability/Brier/log-loss style evaluation.
- Kelly-style sizing and confidence-adjusted risk measures are inputs to allocation governance rather than standalone position-size commands.

7. Portfolio risk and capital allocation

- Portfolio Greeks, beta-weighted directional exposure, strategy/symbol/sector concentration and correlation are aggregated at portfolio level.
- Risk budgets define capital, heat and concentration constraints before candidate selection.
- Portfolio fit and opportunity cost measure how much incremental value a candidate adds relative to existing positions and alternatives.
- Optimization selects a feasible candidate set under capital, symbol, sector, strategy and correlation constraints; publication of current allocation is a governed authority.

8. Execution and tradability analytics

- Executable bid/ask and strategy package prices are distinguished from theoretical or stale marks.
- Freshness and adverse-drift checks determine whether an approved trade remains executable.
- IBKR market-rule/minimum-tick normalization ensures submitted prices are broker-valid.
- Working-order analytics track age, reprices, envelope consumption, cancellation, terminal broker truth and controlled retries.
- Execution Intelligence evaluates slippage, fill quality and whether realized trading costs were consistent with assumptions.

9. Position-management computations

- Open positions are continuously marked and evaluated against stop, target, time, structural and risk conditions.
- Multi-leg positions are managed as strategy-level objects; the whole position must be closed before the earliest leg expiration under the enforced expiration guardrail.
- Management-quality analytics separate thesis quality from execution and exit quality.
- Portfolio and autonomous-management services reconcile broker truth before acting on local state.

10. M78 Setup Intelligence mathematics

- A setup archetype is a statistically distinct market configuration preserved as its own research population.
- Setup lifecycle is stateful: setup/watch -> confirmation -> retest/continuation or failure/reversal. Only actual stage changes create transition records.
- Setup outcomes use point-in-time detection followed by later matured labels; future outcome fields are excluded from detection.
- Hierarchical empirical probability starts with setup-level priors, then conditions on market, gamma, sector and volatility regimes. Sparse cells are shrunk toward broader priors.
- Readiness gates require minimum observations, positive/negative outcomes and distinct dates. The observed defaults are 100 setup-prior observations, 25 positive, 25 negative and 20 distinct dates.
- Expected-value assessment includes expected return, expected R, capital efficiency, time efficiency and quality/confidence-adjusted utility.
- Prospective certification evaluates frozen shadow predictions against later realized outcomes. M78 has no production-promotion action.

11. Authority time and lineage semantics

- snapshot_timestamp/published_at indicate when computation or publication occurred; they do not necessarily equal the market session represented by the data.
- Effective market date is derived from explicit lineage when available. The repaired M77.40 path resolves portfolio publication -> optimization snapshot -> stock_scanner_run_id -> Stock Scanner lineage.market_as_of_date.
- Freshness, market-session identity and authority status are separate checks and should not be conflated.

12. Governance concepts

- READY means an authority is usable under its contract; DEGRADED means available with explicit limitations; INSUFFICIENT_EVIDENCE is a valid research state; BLOCK/REJECT means downstream action is disallowed.
- Historical performance does not automatically become production authority. Prospective evidence and explicit governance are required.
- Production and research namespaces are deliberately separated so experiments cannot silently alter decision thresholds, portfolio rules, execution or management.

13. Core formulas and interpretations

Concept	Representative form	Interpretation
Expected value	EV = P(win)*AvgWin - P(loss)*AvgLoss - costs	Probability-weighted economic value, not just win rate.
Expected R	E[R] = sum(probability_i * R_i)	Expected payoff normalized by governed risk unit.
Capital efficiency	Expected value / capital at risk	Compares opportunity value per unit of committed capital.
Time efficiency	Expected value / expected holding time	Compares opportunity value per unit of time.
Beta-weighted exposure	sum(position delta * underlying beta * notional scaling)	Directional portfolio exposure in a common benchmark frame.
Correlation concentration	weighted pairwise exposure correlation	Measures diversification versus common-factor crowding.
Calibration	predicted probability vs realized frequency	Tests whether stated probabilities correspond to observed outcomes.